

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

Electronics and Communication Engineering Department

Data Structures and Algorithms (IIPC 210)

Mid Sem Exam-I

Max. Marks: 15

Duration: 50 min.

All questions are compulsory.

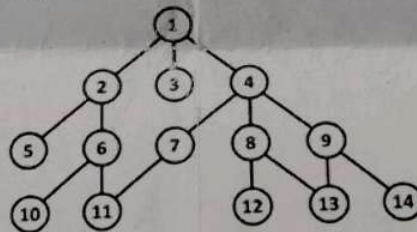
Q1. The recurrence relation for a divide-and-conquer algorithm is given as:

$$T(n) = 3T(n/2) + O(n)$$

Use the **substitution method** to determine the asymptotic complexity of $T(n)$. Clearly show the expansion and general form before concluding the final complexity.

Q2. Given the following **undirected graph**:

- (a) Perform **BFS traversal** starting from node 1.
- (b) Perform **DFS traversal** starting from node 1.



Q3. Solve the following graph using both **Prim's** and **Kruskal's Algorithms**.

